Project Interim Report

On

File Compression Simulator
(Using Core Java & OOP Concepts)

RU-100-01-00012

Object Oriented Programming with Java

SESSION: 2025-26

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Abstract

This project focuses on the design and implementation of a Digital Wallet Payment System using Java and Object-Oriented Programming (OOP) concepts. A digital wallet allows users to store money electronically and perform secure transactions such as deposits and fund transfers.

The need for such a system arises due to the increasing demand for cashless and secure payment methods in today's digital world. This project simulates real-world wallet operations while ensuring data security and transaction reliability.

The system uses encapsulation to protect user balance, static variables to manage cashback rates, and exception handling to deal with errors like insufficient balance or invalid transactions. It provides features such as adding money, transferring funds between wallets, and checking balance.

Overall, this project demonstrates how OOP concepts can be applied to build a simple, secure, and efficient financial application

Introduction

In the modern digital era, online payment systems have become essential for everyday transactions. Digital wallets provide a convenient and secure way to handle money without using physical cash.
This project implements a Digital Wallet Payment System using Java. The system allows users to deposit money, transfer funds between wallets, and receive cashback on transactions.
The project is built using core OOP principles:

- Encapsulation ensures secure access to wallet balance
- Static variables allow shared cashback policies
- Exception handling ensures safe and error-free transactions

Features of the Project

- Secure balance management
- Wallet-to-wallet money transfer
- Cashback system
- Error handling for invalid operations

Objectives

- To implement a secure digital wallet system
- To apply OOP concepts like encapsulation and abstraction
- To enable safe fund transfer between users
- To reduce manual cash transactions
- To handle runtime errors using exception handling

Scope

This project is suitable for learning and small-scale financial simulations. It can be extended in future with:

- Database integration
- User authentication (login system)
- Mobile app interface
- Online payment gateway integration

System Requirements

Hardware

- Computer/Laptop with minimum 4GB RAM

Software

- Java JDK
- IDE (VS Code / Eclipse / IntelliJ)

Methodology

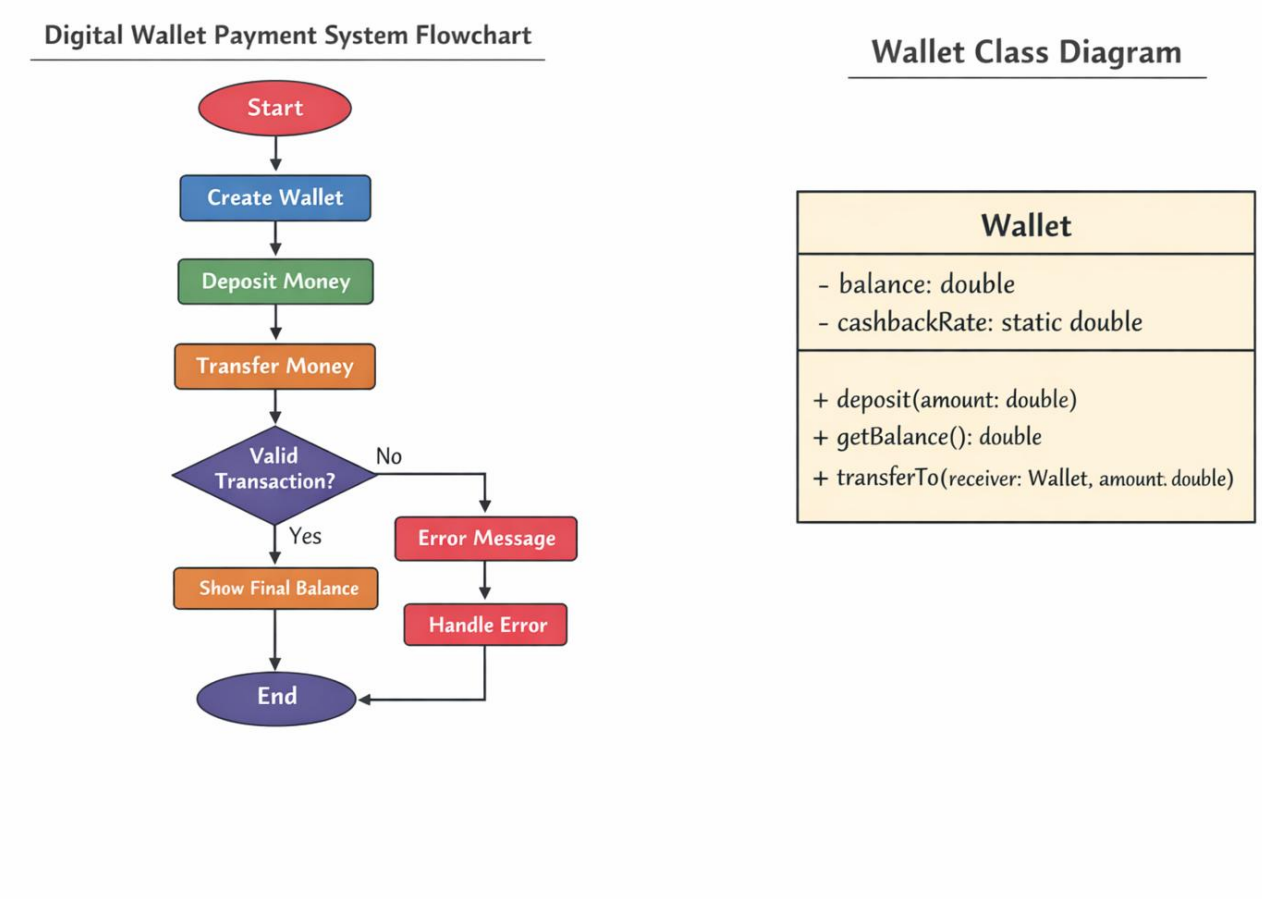
The system works in the following steps:

1. User creates wallet objects
2. User deposits money into wallet
3. User selects transfer option
4. System checks:
    - Valid amount
    - Sufficient balance
5. If valid:
    - Amount is transferred
    - Cashback is added
6. If invalid:
    - Error message is displayed
7. Final balance is shown

Flowchart

Includes Flowchart (Start, Menu, Choice, Operations) , and Class Diagram representing system structure.

Sample class diagram –



ClassName: Wallet

Variables:

- private double balance
- static double cashbackRate

Methods:

- deposit(double amount)
- getBalance()

- transferTo(Wallet receiver, double amount)

Implementation

The project mainly uses one class:

Wallet Class

- Stores balance as a private variable (security)
- Uses a static cashback rate shared by all wallets
- Deposit method adds money
- Transfer method:
  - Checks balance
  - Transfers money
  - Adds cashback
  - Handles errors using try-catch

Main Class

- Creates wallet objects
- Performs operations like deposit and transfer
- Displays results

Sample Input/Output

Initial Balance (W1): 1000
Transferring 300...

Transaction successful. Cashback: 6.0

Final Balance (W1): 706.0
Final Balance (W2): 300.0

Future Enhancements

- Add login and user authentication
- Connect with database (MySQL)
- Mobile app development
- Add bill payment features
- Add transaction history

Gantt Chart (Write like this)

Task	Duration
Planning	2 Days
Design	2 Days
Coding	4 Days
Testing	2 Days
Documentation	2 Days

Conclusion

The Digital Wallet Payment System successfully demonstrates the use of Java OOP concepts such as encapsulation, static variables, and exception handling. The project provides a simple and secure way to simulate digital transactions. It also helps in understanding real-world financial system design.

References

[1] H. Schildt, *Java: The Complete Reference*, 11th ed.
[2] Oracle, "Java Documentation." Available: https://docs.oracle.com
[3] TutorialsPoint Java Guide